

Xin HUANG

Hong Kong

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HKU graduate in Mathematics and Computer Science with a strong foundation in optimization and data analysis. Gained experience in research and software development, and proficient in Python and JavaScript. Eager to apply academic knowledge and practical skills in technical roles.

Education

University of Hong Kong,

Hong Kong *September 2021 - July 2025*

B.Sc. (Mathematics and Computer Science)

Dean's Honours List in faculty of Science, HKU

Cumulative GPA: 3.67/4.3

Expected graduation time: July 2025

Course experience:

Mathematics: Operations Research, Probability and Statistics, Scientific Computing, Differential Equations, Optimization, Numerical analysis, Financial Calculus, Game Theory, Functional Analysis, Linear algebra I & II;

Computer Science: Algorithm Design, Data structure and algorithm, Artificial Intelligence, Database Management

Experience

Research Assistant, *University of Hong Kong,*
HKU Business School

Hong Kong *November 2024 - March 2025*

- Analyzed mathematical problems and proved operations management models, optimizing contracts and strategies under constraints
- Applied Linear Programming and optimization techniques to solve resource allocation and supply chain issues
- Strengthened skills in mathematical modeling and data-driven decision-making.

Research Intern, *University of Alberta,*
Department of Mathematical and Statistical Science
Experiments with graph signal processing

Alberta, Canada *June 2024 - August 2024*

- Conducted data analysis and experimental evaluations on the open-sourced published data to examine the feasibility of applying Fourier and Wavelet analysis techniques
- Investigated the integration of the Kolmogorov-Arnold Network (KAN) with collaborative filtering in recommender systems, enhancing both efficiency and performance by replacing the original B-spline or Fourier bases with Wavelet basis
- Explored the possibility of using hypergraphs with the KAN architecture to model interactive relationships between users and products, investigating for the development of an improved recommendation system that outperformed classical recommendation system models including NGCF and LightGCN.

Student Teaching Assistant, *University of Hong Kong,*
MATH1853 Linear Algebra, Probability and Statistics & COMP1117 Computer Programming

Hong Kong *September 2023 - April 2025*

- Prepared auxiliary teaching materials to support lecturer instruction
- Generated sets of topic-related questions to enhance student understanding
- Provided 'coding clinic' sessions (consultation) for students to help students addressing problems they encountered.

Research Project

Efficient numerical schemes for Path-dependent PDEs

February 2025 - May 2025

- Studied the efficient numerical algorithms for Path-dependent Partial Differential Equations in non-Markov process modeling to price financial derivatives
- Conducted readings and duplication on the existing extended theorems on Path-dependent PDE with monotone schemes for fully nonlinear PPDE, including finite difference method and probabilistic scheme.

Activities

Front-End Engineer Volunteer, SAGA, Beyond the Pivot (BTP)
IT Department

Hong Kong July 2024 - Present

- SAGA, a volunteering project aiming to provide free lessons to leukemia children
- Developed and maintained front-end components for the SAGA Web pages, including team members showcase, volunteer application, cooperator information and souvenir exchange pages.
- Improved user interface experience by adding animations and dynamic effects into webpage.

Achievement

- Dean's Honours List: Top 10% in the class of Bachelor of Science. (2022-2023, 2023-2024)
- The University of Hong Kong Full Scholarship for Outstanding Academic Talents (2021-2022, 2022-2023, 2023-2024, 2024-2025)

Skills

- **Programming:** Python, JavaScript, C++, CSS, HTML, Vue.js, SQL, Tableau
- **Language:** English (IELTS 8), Mandarin, Cantonese